KRIBU ENGINEERING ARCHITECTURE

High-Level Architecture and LangGraph Workflow

A unified, industry-neutral explanation of how data enters KRIBU, how authorization constrains retrieval, how the AI Engine reasons over evidence, and how approved actions return to client systems.

> **Core rule:** relevance and authorization are separate. KRIBU Guard decides what a user may access; KRIBU Brain searches only inside that authorized scope; KRIBU Engine reasons only over the resulting evidence.

KRIBU component · LangGraph subgraph · External actor/system · Data store · Decision or security gate

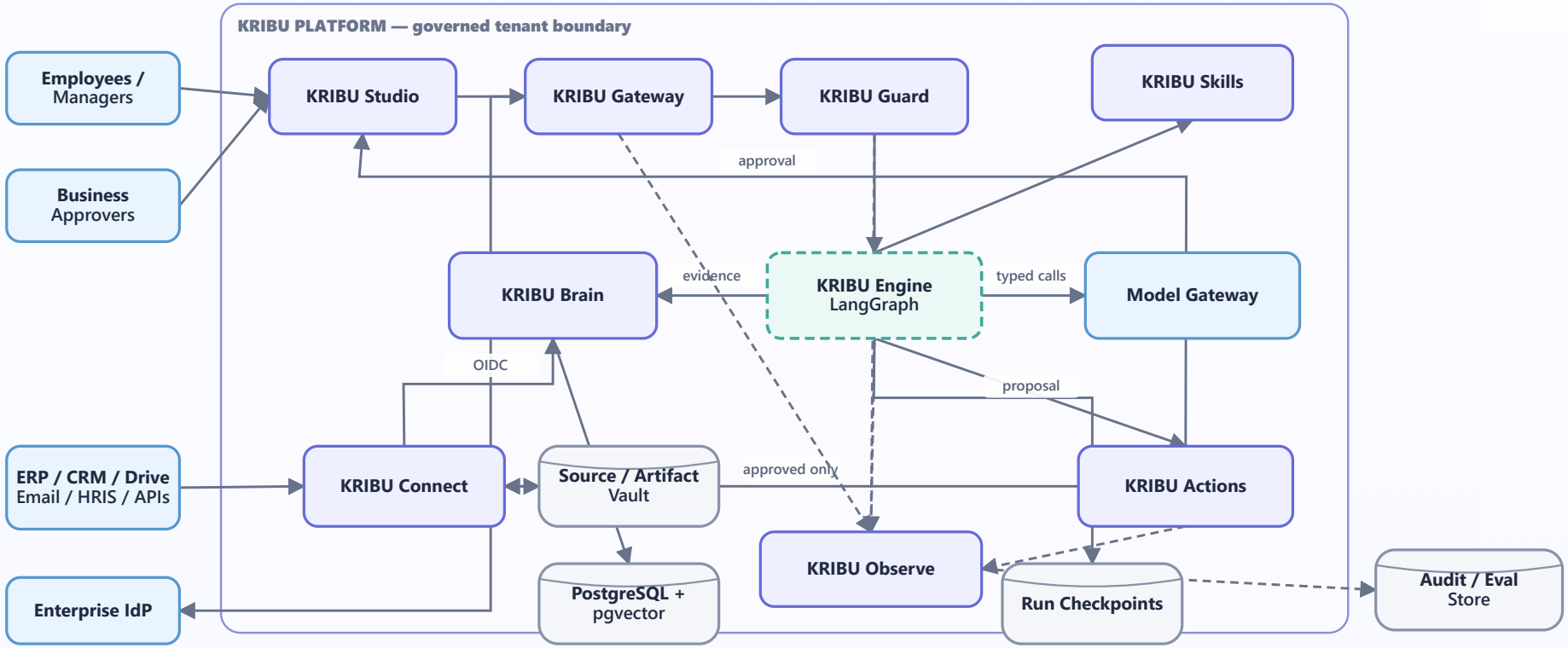
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Prepared for KRIBU · 16 October 2026 · Architecture target, not a claim of implemented functionality

FIGURE 1

Full KRIBU Platform Dataflow

The stable product components and the major synchronous, asynchronous, data, approval and audit flows.



Front door. Users work in Studio. Gateway authenticates, rate-limits and streams. Guard creates an authorization scope before Engine or Brain can access business data.

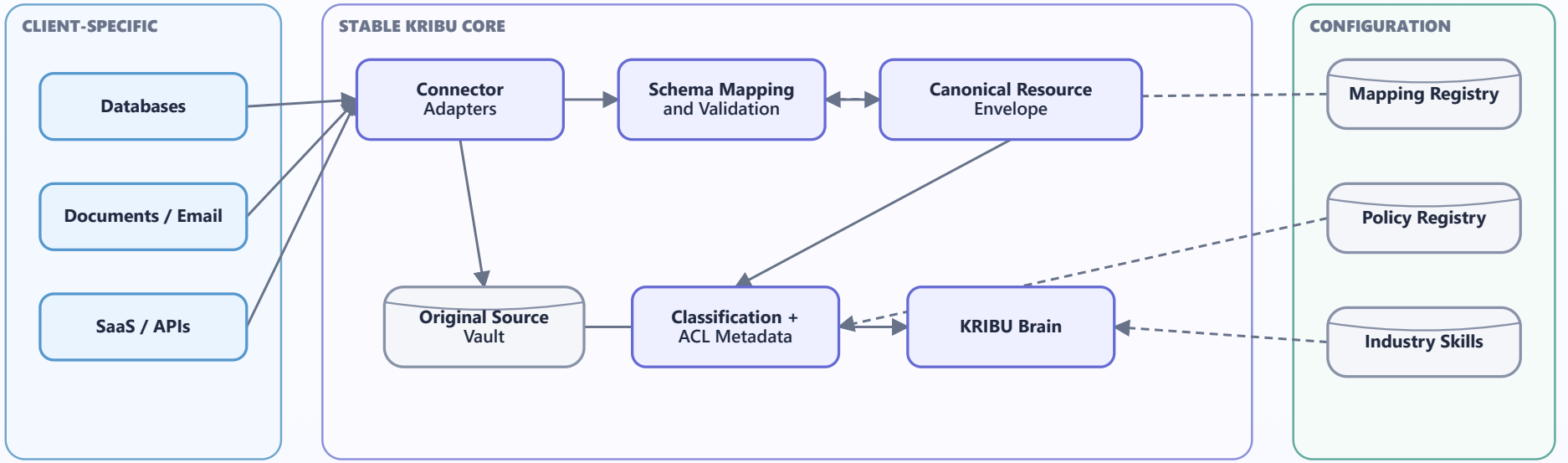
Intelligence path. Engine loads a versioned Skill, requests permission-filtered evidence from Brain, calls models through the Model Gateway, and validates citations.

Controlled action. Side effects move through Actions and human approval before Connect can call an external system. Observe receives audit events throughout.

FIGURE 2

Industry-Neutral Data Adaptation

Different database schemas are absorbed by adapters and mappings; core KRIBU contracts remain stable.



Preserve source truth. Original files and records remain versioned in the Source Vault. Mapping never destroys the source representation.

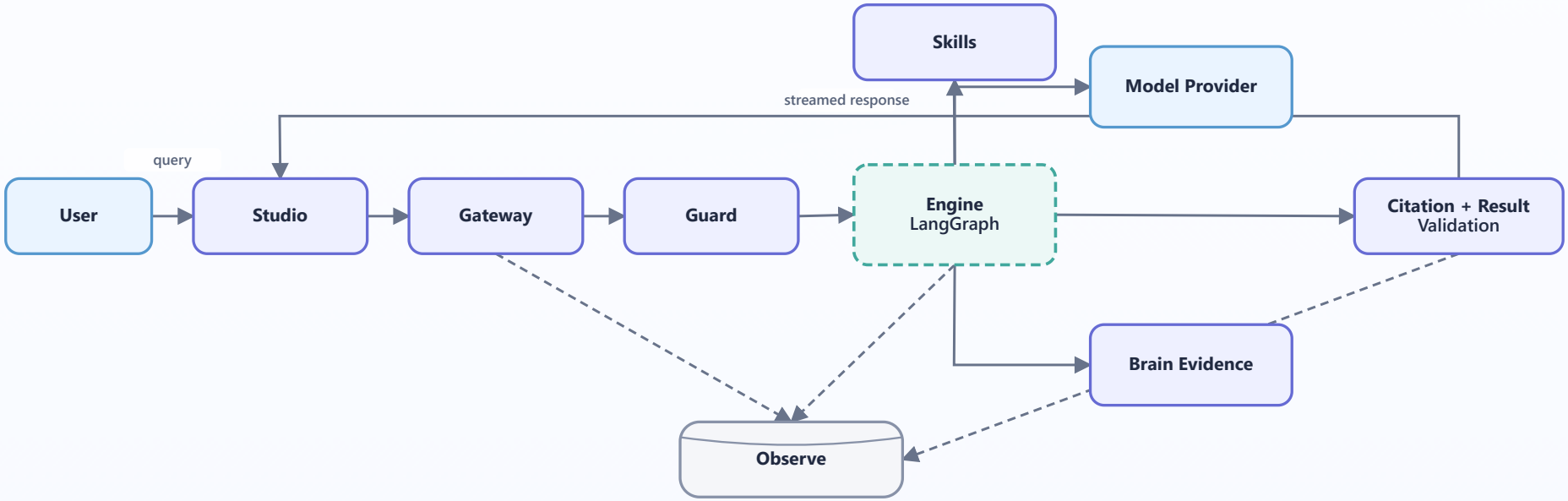
Canonical envelope. Every item receives tenant, source, resource type, timestamps, classification, ACL, relationships and provenance—even when its business fields differ.

Extend without rebuilding. A new industry adds connector adapters, schema mappings, policies and Skills. It does not replace Brain or Engine.

FIGURE 3

User Query to User Response

The simplest reading path through the platform for an evidence-backed question.



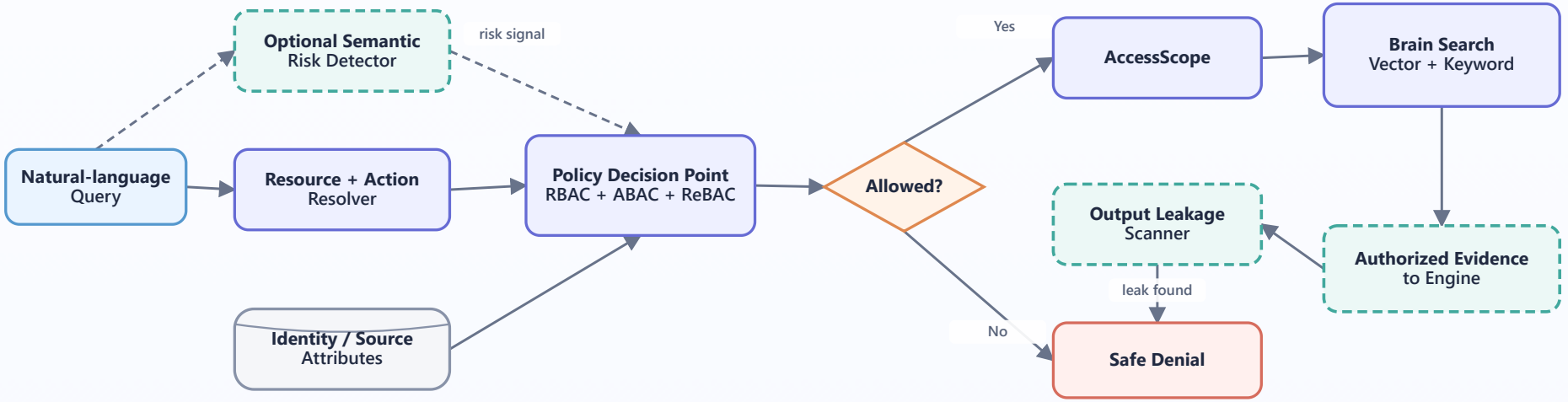
Request phase. Studio sends the query to Gateway. Guard resolves identity, tenant, roles, attributes and document scope. Engine then chooses an approved Skill and requests evidence.

Response phase. Engine interprets authorized evidence, performs deterministic calculations, validates every material citation, stores the result, and streams a safe response through Gateway.

FIGURE 4

Authorization-Aware Retrieval

Policy decides access. Vector and keyword search decide relevance only inside the allowed scope.



Dynamic policies. Guard evaluates principal, action, resource, relationship, environment and policy. Rules describe resource classes—not named people.

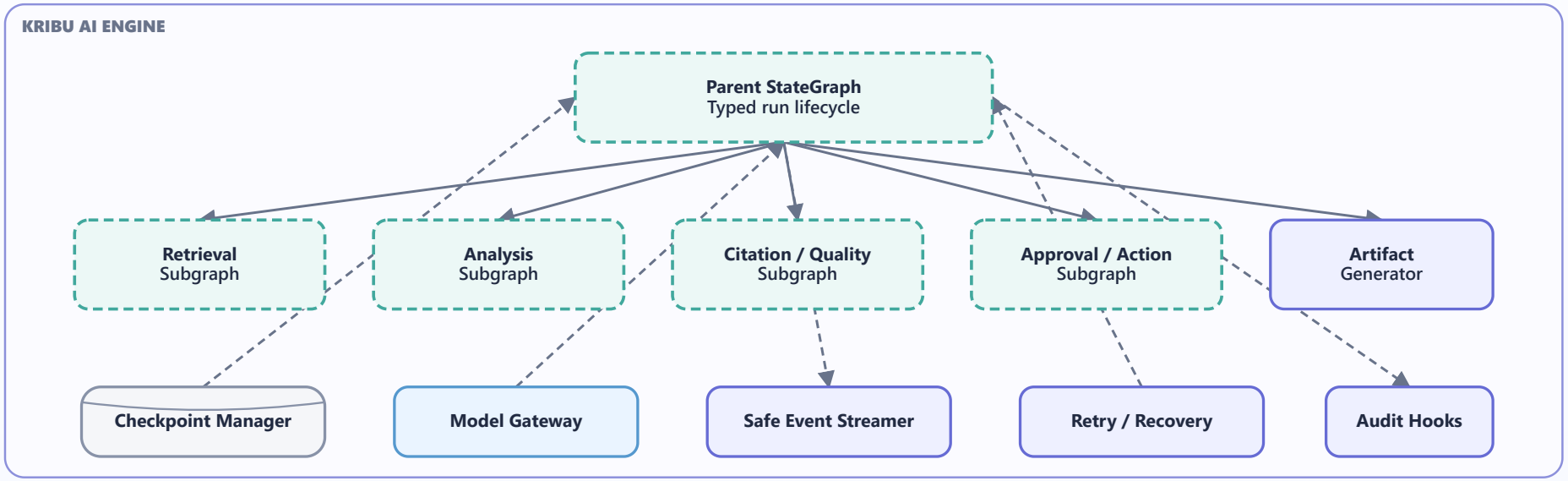
Optional semantic detector. A lightweight model can recognize synonyms, prompt injection and possible leakage. It returns a risk signal; it cannot grant permission.

Security invariant. Unauthorized chunks never reach the reasoning model. Retrieval filters and PostgreSQL RLS enforce the decision again.

FIGURE 5

Inside the KRIBU AI Engine

A controlled orchestration runtime rather than a single unrestricted autonomous agent.



Parent graph. Owns the run lifecycle and routes typed state between governance, retrieval, analysis, validation, artifact and action stages.

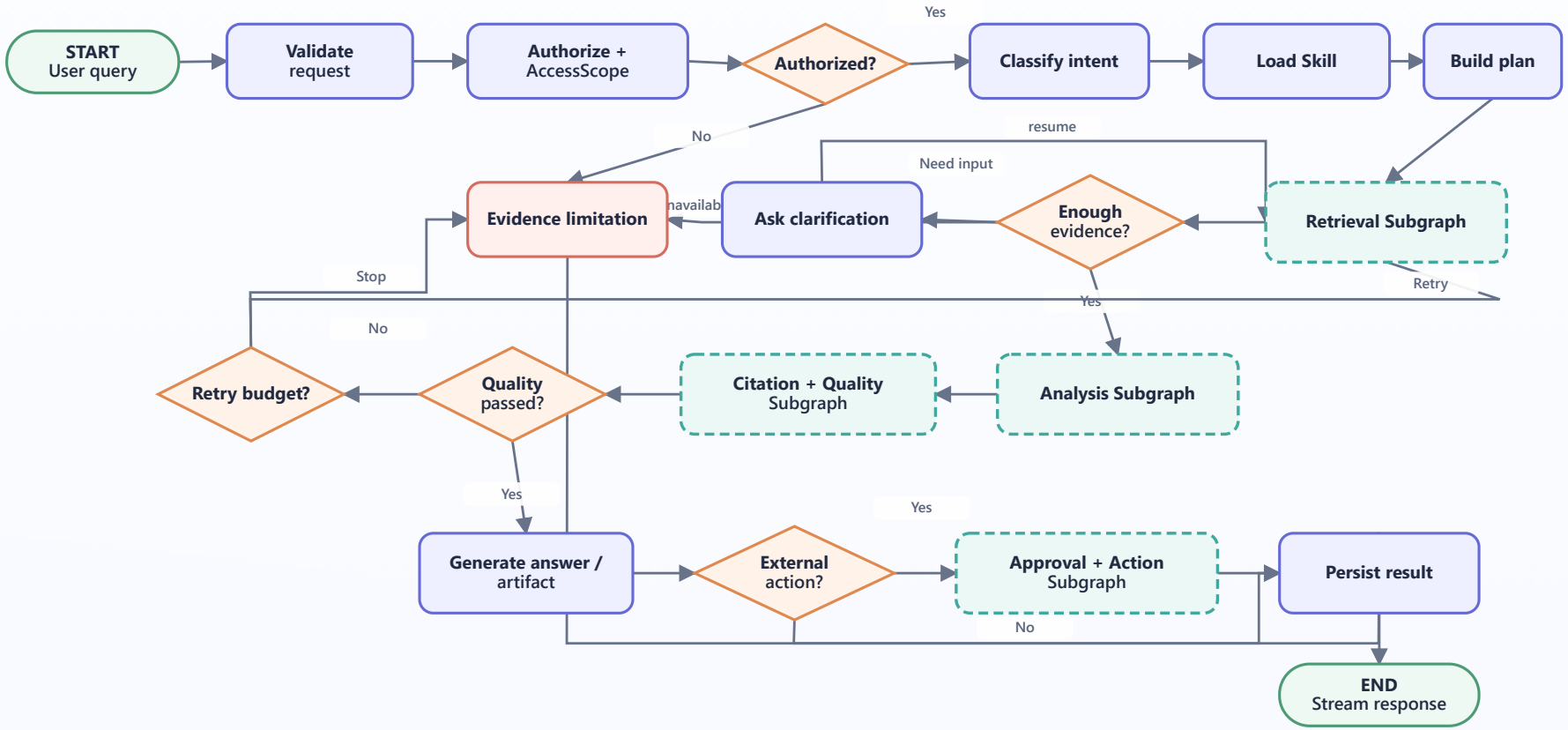
Subgraphs. Reusable workflows expose typed inputs and outputs. Industry Skills configure them without changing Engine infrastructure.

Runtime services. Checkpoints, model adapters, safe event streaming, retries and audit hooks support long-running and resumable execution.

FIGURE 6

Parent LangGraph Workflow

The complete run from validated request to final answer or approved external action.



Deterministic gates. Request validation, authorization, calculations, citation integrity, retry limits and action policy are code-driven.

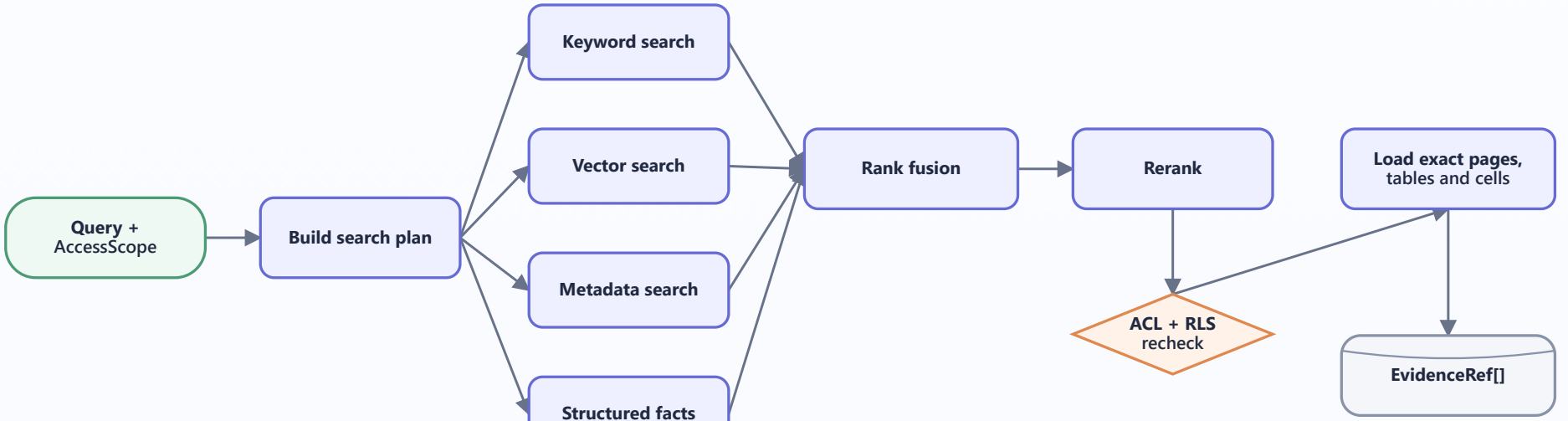
Model-assisted stages. Intent classification, extraction and synthesis use typed model outputs and cannot override Guard.

Safe exits. Missing evidence returns a limitation. Sensitive actions pause for approval. Rejection and failure remain valid audited outcomes.

FIGURE 7

Retrieval Subgraph

Parallel retrieval modes are fused, reranked, permission-checked and expanded into exact evidence.



Parallel discovery. Keyword, vector, metadata and structured-fact searches answer different aspects of the query.

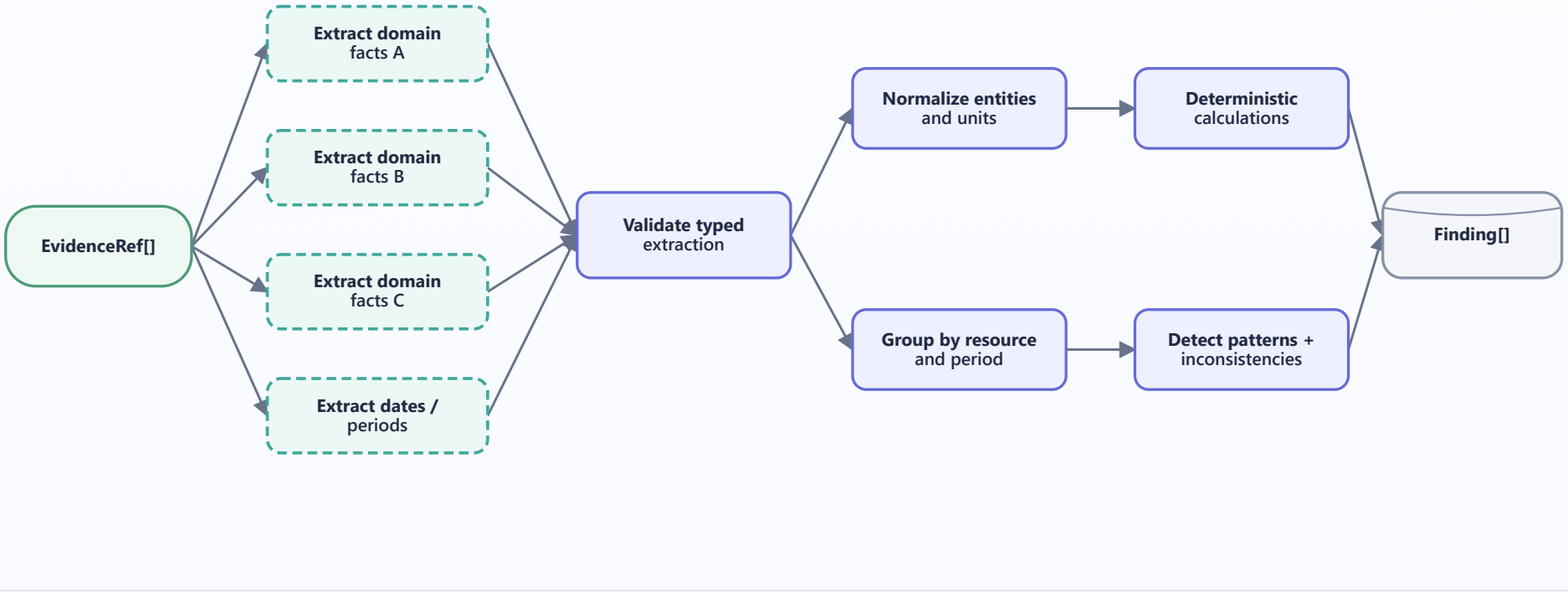
Permission preservation. AccessScope is applied during the query and rechecked before exact source windows are returned.

Output. The subgraph returns `EvidenceRef` objects containing document version, page/cell, excerpt, score and provenance.

FIGURE 8

Analysis Subgraph

Typed extraction fans out across evidence, then deterministic normalization and comparison produce findings.



Skill-dependent extractors. Project milestones, quotation terms, complaint categories or employee facts are configured by the selected Skill.

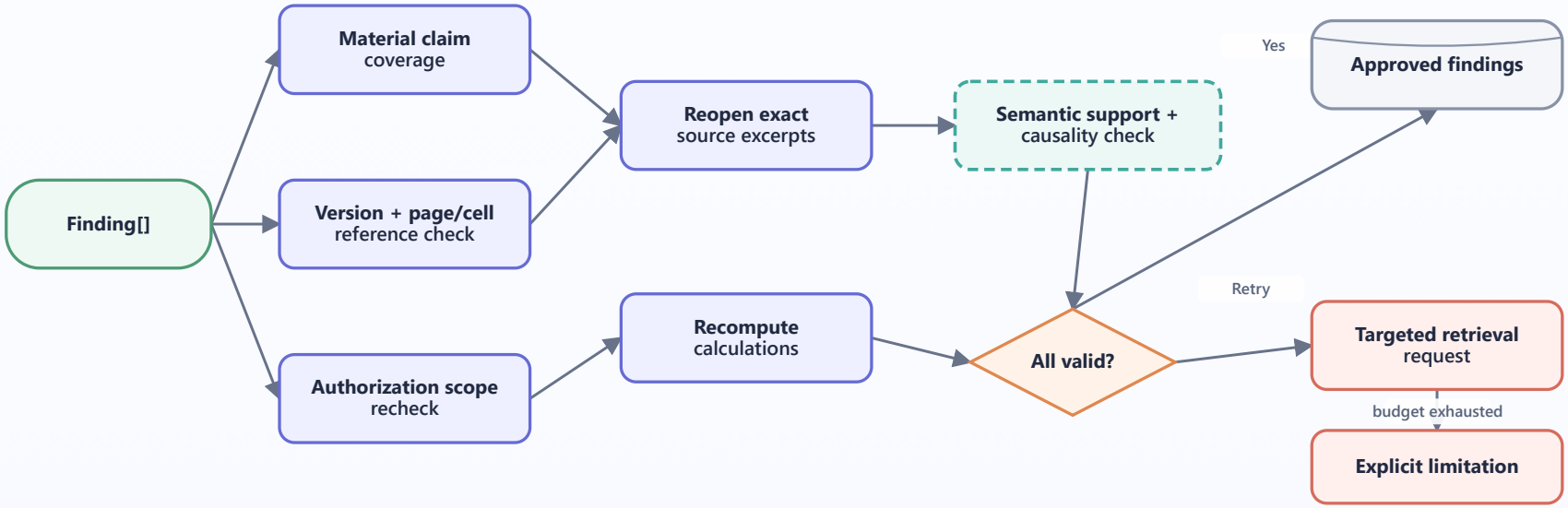
Code owns arithmetic. Variances, dates, totals and thresholds are calculated with deterministic code—not guessed by the model.

Finding contract. Every finding carries category, severity, confidence, limitations, inference level and evidence references.

FIGURE 9

Citation and Quality Subgraph

Before presentation, findings are checked against exact source locations and deterministic calculations.



Mechanical validation. Document versions, ACL scope, page/cell references and calculations are verified in code.

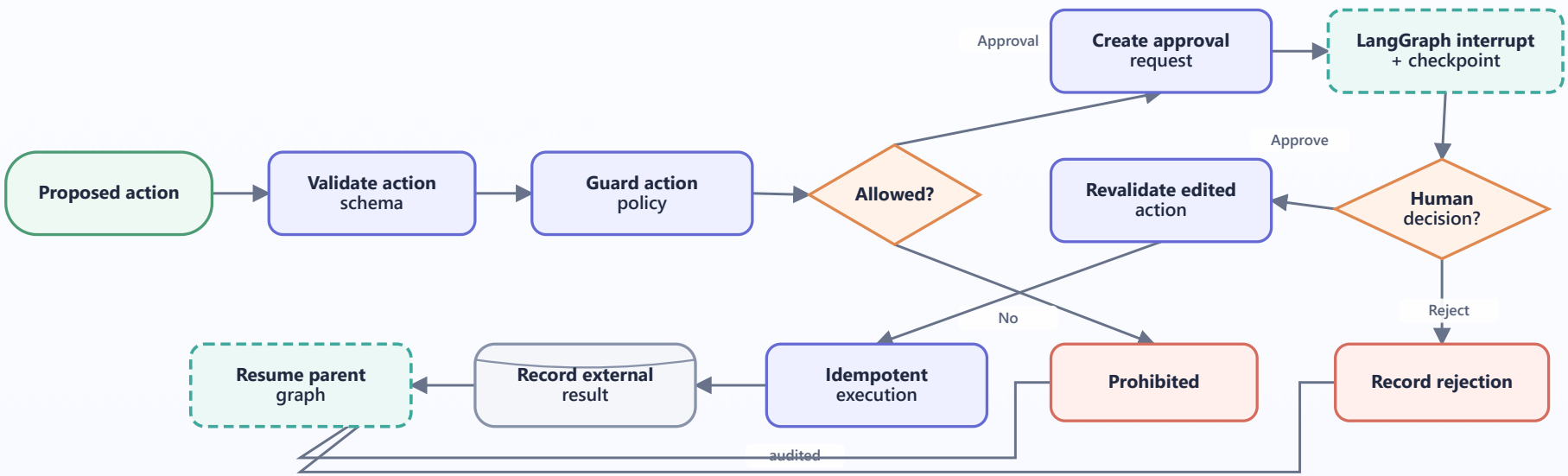
Semantic support. A constrained evaluator checks whether the cited excerpt actually supports the claim and flags unsupported causality.

Repair loop. Failed claims create targeted retrieval requests. After a bounded retry limit, KRIBU returns an explicit limitation.

FIGURE 10

Approval and Action Subgraph

Emails, tasks and record changes remain proposals until policy and human approval permit execution.



Interrupt before effect. LangGraph checkpoints the run and pauses before any external write. The interrupted node may safely resume later.

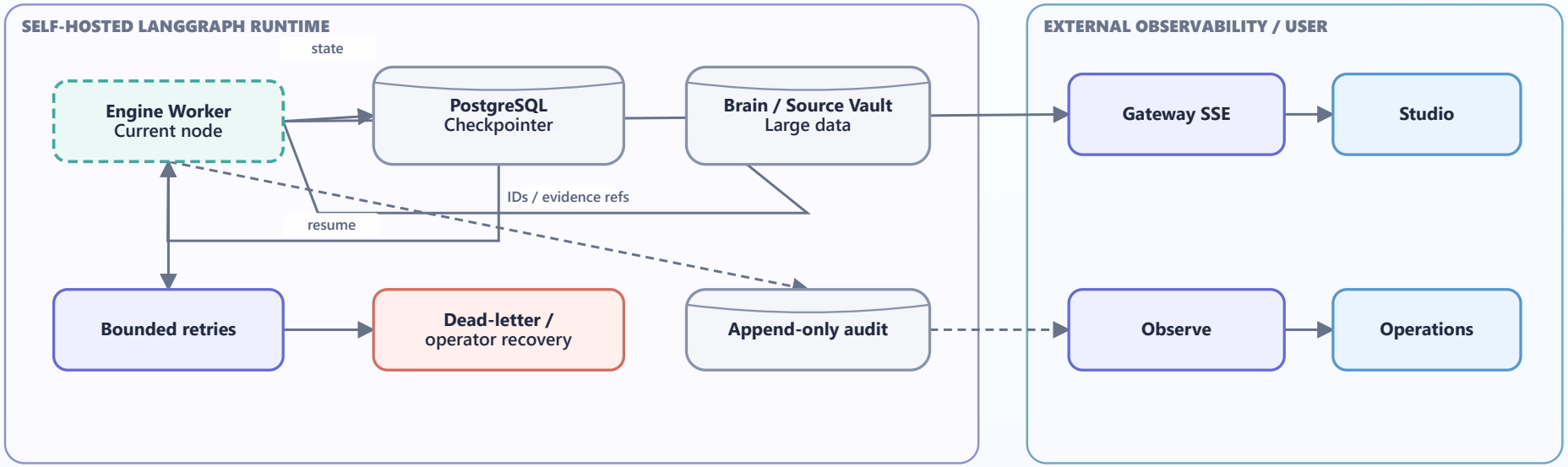
Revalidation. Edited actions are checked again for permission, recipient, classification and schema before execution.

Idempotency. Every outbound action uses a stable idempotency key so retries cannot send or create the same item twice.

FIGURE 11

Persistence, Streaming and Recovery

Graph state survives restarts and long waits; business knowledge remains separate from execution memory.



Checkpoint state. PostgreSQL stores small serializable run state keyed by tenant and run. Large documents stay in Brain and the Source Vault.

User visibility. Gateway streams safe milestones such as retrieval completed, approval requested and run completed—never hidden prompts or unauthorized excerpts.

Failure handling. Transient operations retry with limits; repeated failures enter a dead-letter path and produce an auditable recoverable status.

REFERENCE

Component Responsibilities and Boundaries

The concise contract for reading every figure in this document.

Component	Primary responsibility	Must never do
KRIBU Studio	User workspace, source review, approvals, Skill Studio and administration.	Bypass Gateway or expose hidden Engine state.
KRIBU Gateway	API boundary, tenant context, validation, throttling and safe SSE streaming.	Make domain findings or authorize data by itself.
KRIBU Guard	Identity, RBAC, ABAC/ReBAC, resource classification, AccessScope and action policy.	Use an LLM as the final authorization authority.
KRIBU Connect	Schema-specific adapters, sync, webhooks, uploads and approved outbound actions.	Execute an unapproved side effect or erase source provenance.
KRIBU Brain	Source versions, canonical resources, chunks, facts, relationships, embeddings and provenance.	Return evidence outside Guard's scope.
KRIBU Engine	LangGraph orchestration, evidence analysis, deterministic calculations, validation and artifact generation.	Grant access, invent evidence or directly execute sensitive actions.
KRIBU Skills	Versioned workflow configuration, typed schemas, permissions, prompts, tools and test datasets.	Self-modify or publish without evaluation and review.
KRIBU Actions	Approval requests, decision capture, revalidation and idempotent execution commands.	Treat model intent as approval.
KRIBU Observe	Audit, traces, costs, quality evaluations, operational metrics and exports.	Become the source of business authorization.

Stable core. New industries extend Connector mappings, classification vocabularies, policies and Skills. The core component boundaries and evidence contracts remain unchanged.

Trust boundary. Source documents and external records are untrusted input. They may supply evidence, but they cannot change policies, select permissions or authorize tools.